THE KERNEL OF A MAP ONTO AN I-PROJECTIVE MODULE WITH PROJECTIVE SOURCE IS I-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of the cards already published from this chapter, so that the batch selector treats them as one source instead of deferring them as a same-source clash.

Allowed standard prerequisites (editorial).

- Definition of an I-projective module. introduced in the same section, source lines 17780-17810, before the counted statement. Supplies both the hypothesis on M and the conclusion for K in terms of the annihilation of extension classes by elements of I.
- The class of a short exact sequence in Ext^1 as the image of the identity of the kernel under the connecting map. used at proof lines 17819-17821. Turns the hypothesis on the extension into a statement about the identity map of K, which is the object the argument operates on.
- Annihilation of an extension class by a means that multiplication by a on the kernel factors through the middle term. used at proof lines 17821-17823 to obtain the factorisation a : K to K through K to P. Is the final step that yields I-projectivity of K.

Source reference. Original lemma 0G95: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20290–20290. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let*

$$0 \to K \to P \to M \to 0$$

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be a short exact sequence of R-modules. If M is I-projective and P is projective, then K is I-projective.

Proof. The element $\mathrm{id}_K \in \mathrm{Hom}_R(K, K)$ maps to the class of the given extension in $\mathrm{Ext}^1_R(M, K)$. Since by assumption this class is annihilated by any $a \in I$ we see that $a : K \to K$ factors through $K \to P$ and we conclude. □